



Optional material for the
Text Analytics course

Question Answering

2019–20

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<http://www.aueb.gr/users/ion/>

These slides are partly based on material from the book *Speech and Language Processing* by D. Jurafsky και J.H. Martin, 2ⁿ edition, Pearson Education, 2009, and 3rd edition (draft).

Contents

- **Question answering for document collections and structured data.**
- **Evaluating QA systems, including evaluation measures from information retrieval.**

Show additional filters

Article types
Review
More ...

Text availability
Abstract available
Free full text available
Full text available

Publication dates
5 years
10 years
Custom range...

Species
Humans
Other Animals

Clear all

Show additional filters

Display Settings: Summary, 20 per page, Sorted by Recently Added Send to: Filters: Manage Filters

Did you mean: *what is the methyl donor of dna (cytosine-5) methyltransferases* (34 items)

Results: 1 to 20 of 34 << First < Prev Page 1 of 2 Next > Last >>

- ☐ [Cytosine-to-uracil deamination by SssI DNA methyltransferase.](#)
1. [Stier I, Kiss A.](#)
PLoS One. 2013 Oct 21;8(10):e79003. doi: 10.1371/journal.pone.0079003. eCollection 2013.
PMID: 24205358 [PubMed - in process] **Free PMC Article**
[Related citations](#)
- ☐ [Long-term exposure to cigarette smoke extract induces hypomethylation at the RUNX3 and IGF2-H19 loci in immortalized human urothelial cells.](#)
2. [Chen LM, Nergard JC, Ni L, Rosser CJ, Chai KX.](#)
PLoS One. 2013 May 28;8(5):e65513. doi: 10.1371/journal.pone.0065513. Print 2013.
PMID: 23724145 [PubMed - indexed for MEDLINE] **Free PMC Article**
[Related citations](#)
- ☐ [Diet influences expression of autoimmune-associated genes and disease severity by epigenetic mechanisms in a transgenic mouse model of lupus.](#)
3. [Strickland FM, Hewagama A, Wu A, Sawalha AH, Delaney C, Hoeltzel MF, Yung R, Johnson K, Mickelson B, Richardson BC.](#)
Arthritis Rheum. 2013 Jul;65(7):1872-81. doi: 10.1002/art.37967.
PMID: 23576011 [PubMed - indexed for MEDLINE]
[Related citations](#)
- ☐ [The 'golden age' of DNA methylation in neurodegenerative diseases.](#)
4. [Fuso A.](#)
Clin Chem Lab Med. 2013 Mar 1;51(3):523-34. doi: 10.1515/cclm-2012-0618. Review.
PMID: 23183753 [PubMed - indexed for MEDLINE]
[Related citations](#)
- ☐ [Wilson's disease: changes in methionine metabolism and inflammation affect global DNA methylation in early liver disease](#)
5.

New feature
Try the new Display Settings option - Sort by Relevance

(methyl[All Fields] AND ("tissue donors"[MeSH Terms] OR ("tissue"[All Fields] AND "donors"[All Fields]) OR "tissue donors"[All Fields] OR

Search

See more...

Relevant document snippets

Release of sunflower seed dormancy by cyanide: cross-talk with ethylene signalling pathway

Introduction Cyanide is a compound known to stimulate germination and to release dormancy of seeds of many species (Taylorson and Hendricks, 1973; Roberts and Smith, 1977; Bogatek and Lewak, 1988; Côme et al., 1988; Bethke et al., 2006). Seed dormancy is defined as the property of a seed that prevents its germination in apparently favourable conditions (Finch-Savage and Leubner-Metzger, 2006). Despite the well-described effects of cyanide on germination and dormancy, the cellular bases of its mechanism are poorly understood, and seem moreover to vary from one species to another. Different hypotheses have been proposed to explain the stimulatory effect of cyanide on germination and dormancy (Côme and Corbineau, 1989). ✕ According to Taylorson and Hendricks (1973), the cyanhydric gas could react with L-cysteine to give the β -L-cyanoalanine necessary for the synthesis of arginine and aspartic acids which could be limiting factors for germination. Hagesawa et al. (1994) suggested that this increase in the amino acid pool might also promote germination by decreasing the water potential in embryonic axis. However, other respiratory inhibitors which are not metabolized, such as NaN_3 or Na_2S , have the same effect as KCN in various species (Roberts and Smith, 1977; Côme and Corbineau, 1989). Some studies proposed that the beneficial effect of cyanide on germination might involve the cyanide-insensitive pathway (Esashi et al., 1979, 1981b; Upadhyaya et al., 1983), the pentose phosphate pathway (Roberts and Smith, 1977; Côme and Corbineau, 1989), the glycolysis (Bogatek, 1995) or the hydrolysis of oligosaccharides and their catabolism (Bogatek and Lewak, 1991; Bogatek et al., 1999). Cyanide is also known to interact with reactive oxygen species (ROS) metabolism; it is an inhibitor of Cu/Zn superoxide dismutase (SOD) (Bowler et al., 1992) and catalase (CAT) (Tejera García et al., 2007) and it has been demonstrated to induce oxidative stress and lipid peroxidation in animals (Johnson et al., 1987; Gunasekar et al., 1998). Oracz et al. (2007) recently demonstrated that cyanide could trigger protein oxidation during sunflower seed dormancy alleviation. At last, cyanide might also interplay with the ethylene signalling pathway. Indeed, hydrogen cyanide is a co-product of ACC oxidase, which converts ACC to ethylene (Peiser et al., 1984), and it has been proposed to stimulate ethylene biosynthesis via a feedback effect (Pirrung and Brauman, 1987). Thus Smith and Arteca (2000) demonstrated that the ACC synthase gene ACS6 was activated by cyanide in Arabidopsis. However, it is actually not known whether ethylene and cyanide share some molecular components of their downstream transduction pathways. The putative relationship between cyanide and ethylene signalling pathways might be particularly relevant for sunflower seeds, whose dormancy is broken by ethylene (Corbineau et al., 1990). The inability of freshly harvested sunflower seeds to germinate at temperatures below c. 15 °C results from an embryo dormancy which is gradually eliminated during dry storage (Corbineau et al., 1990; Corbineau and Côme, 2003). Embryo dormancy is characterized by the inhibition of radicle extension thus preventing excised embryos to grow (Finch-Savage and Leubner-Metzger, 2006). Little attention has been paid to the

Questions of biomedical experts

Yes/no question

“Are there any DNMT3 proteins present in plants?”

‘Exact’ answer

“Yes”

‘Ideal’ answer

“Yes. The plant DOMAINS REARRANGED METHYLTRANSFERASE2 (DRM2) is a homolog of the mammalian de novo methyltransferase DNMT3. DRM2 contains a novel arrangement of the motifs required for DNA methyltransferase catalytic activity.”

Questions of biomedical experts

Factoid question

“What is the methyl donor of DNA (cytosine-5)-methyltransferases?”

Exact answer

“S-adenosyl-L-methionine”

Summary answer

“S-adenosyl-L-methionine (AdoMet, SAM) is the methyl donor of DNA (cytosine-5)-methyltransferases. DNA (cytosine-5)-methyltransferases catalyze the transfer of a methyl group from S-adenosyl-L-methionine to the C-5 position of cytosine residues in DNA.”

Questions of biomedical experts

List question

“Which species may be used for the biotechnological production of itaconic acid?”

Exact answer

“Aspergillus terreus”, “Aspergillus niger”, “Ustilago maydis”

Summary answer

“In 1955, the production of itaconic acid was firstly described for Ustilago maydis. Some Aspergillus species, like A. itaconicus and A. terreus, show the ability to synthesize this organic acid and A. terreus can secrete significant amounts to the media. Itaconic acid is mainly supplied by biotechnological processes with the fungus Aspergillus terreus. Cloning of the cadA gene into the citric acid producing fungus A. niger showed that it is possible to produce itaconic acid also in a different host organism.”

Questions of biomedical experts

Summary-only question

“How do histone methyltransferases cause histone modification?”

‘Exact’ Answer

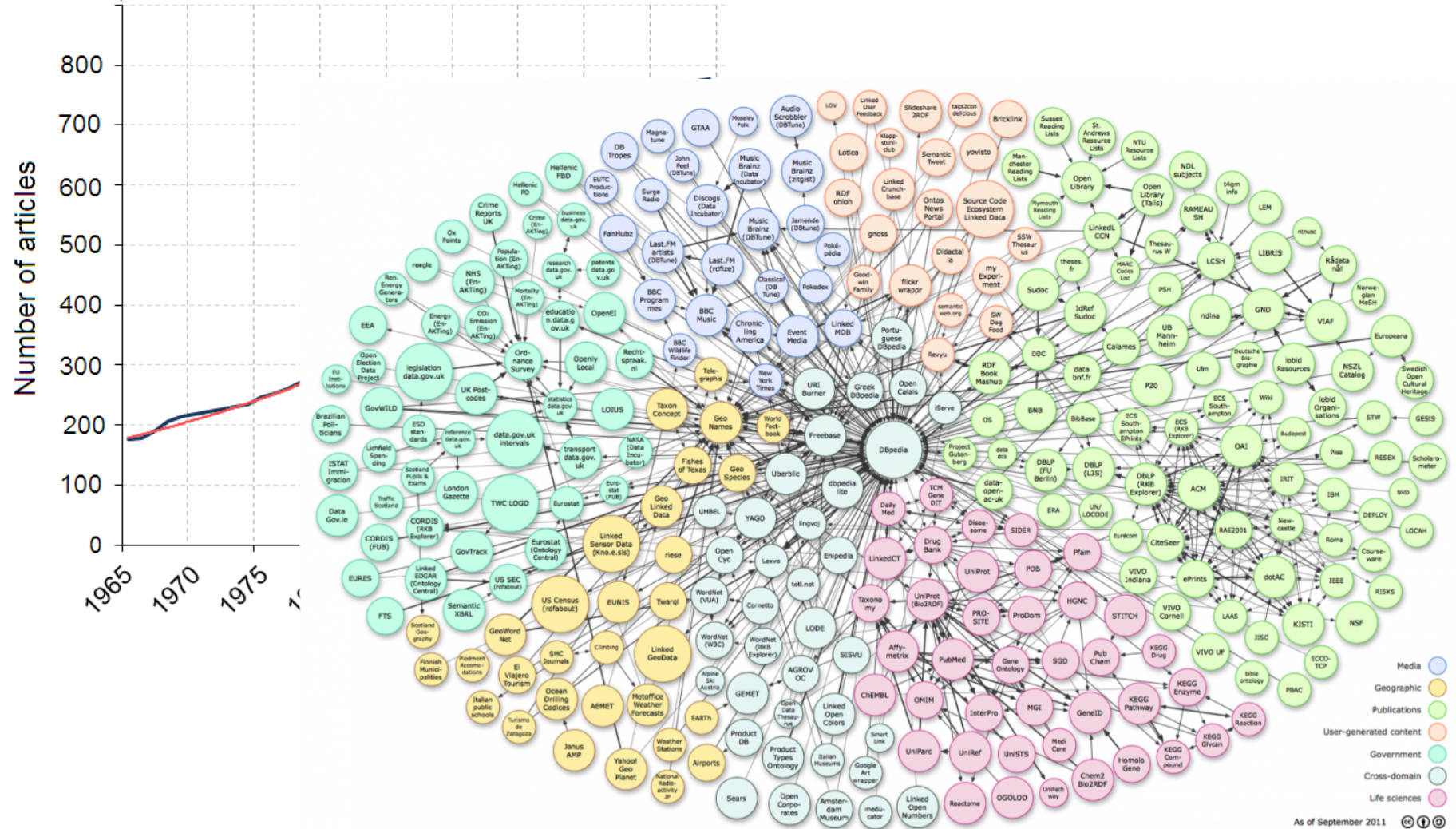
(None)

‘Ideal’ Answer

“Histone methyltransferases (HMTs) are responsible for the site-specific addition of covalent modifications on the histone tails, which serve as markers for the recruitment of chromatin organization complexes. There are two major types of HMTs: histone-lysine N-Methyltransferases and histone-arginine N-methyltransferases. The former methylate specific lysine (K) residues such as 4, 9, 27, 36, and 79 on histone H3 and residue 20 on histone H4. The latter methylate arginine (R) residues such as 2, 8, 17, and 26 on histone H3 and residue 3 on histone H4. Depending on what residue is modified and the degree of methylation (mono-, di- and tri-methylation), lysine methylation of histones is linked to either transcriptionally active or silent chromatin.”

Not only texts: ontologies, linked data, ...

Thousands,



Information from structured data

List question: *“Which forms of cancer is the Tpl2 gene associated with?”*

- Related **RDF triple**:
 - Subject: <http://www4.wiwiss.fu-berlin.de/diseasome/resource/diseases/3003> (lung cancer)
 - Predicate: <http://www4.wiwiss.fu-berlin.de/diseasome/resource/diseasome/associatedGene>
 - Object: <http://www4.wiwiss.fu-berlin.de/diseasome/resource/genes/TPL2>
- Related **concepts**:
 - <http://www.disease-ontology.org/api/metadata/DOID:162> (cancer)
 - http://www.uniprot.org/uniprot/M3K8_RAT (TPL2 synonym)

QA for document collections

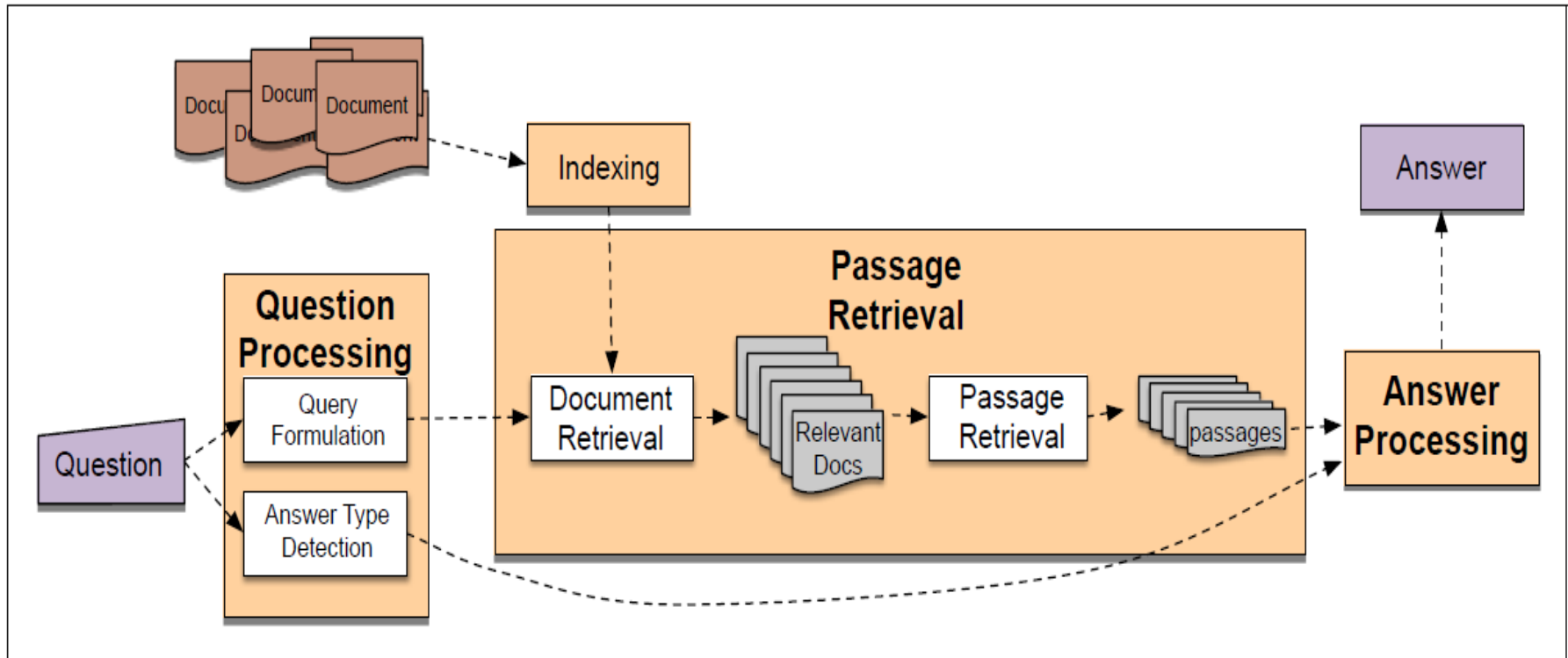


Figure 28.2 IR-based factoid question answering has three stages: question processing, passage retrieval, and answer processing.

Figure from D. Jurafsky & J.H. Martin's "Speech and Language Processing", 3rd Edition.

Question processing

- **Question type and focus:**
 - Is it a **factoid** question asking for **person, location, date, ...?**
And what is the **focus** (e.g., “In which city did Mozart...?”), i.e., roughly the part **to be replaced** by the answer?
 - Is it a **definition/description** question (e.g., “What is thalassemia”) or a **why/how** question (“Why/how do I...”)?
More difficult, often require **generating a summary**.
 - For **question type: classification** problem.
 - Traditionally with **features** like **question words** (e.g., “Who”, “When”), ***n*-grams** (e.g., “was born”), **headword** (“Which English city....?”), **hypernyms of headword** (e.g., city → location), **named entity types** (e.g., does the question contain a location name?).
 - More recently: **word embeddings, MLPs, RNNs, CNNs, BERT...**
 - For **focus: sequence labeling** problem (e.g., methods like those discussed for named entity recognition).

A taxonomy of question types

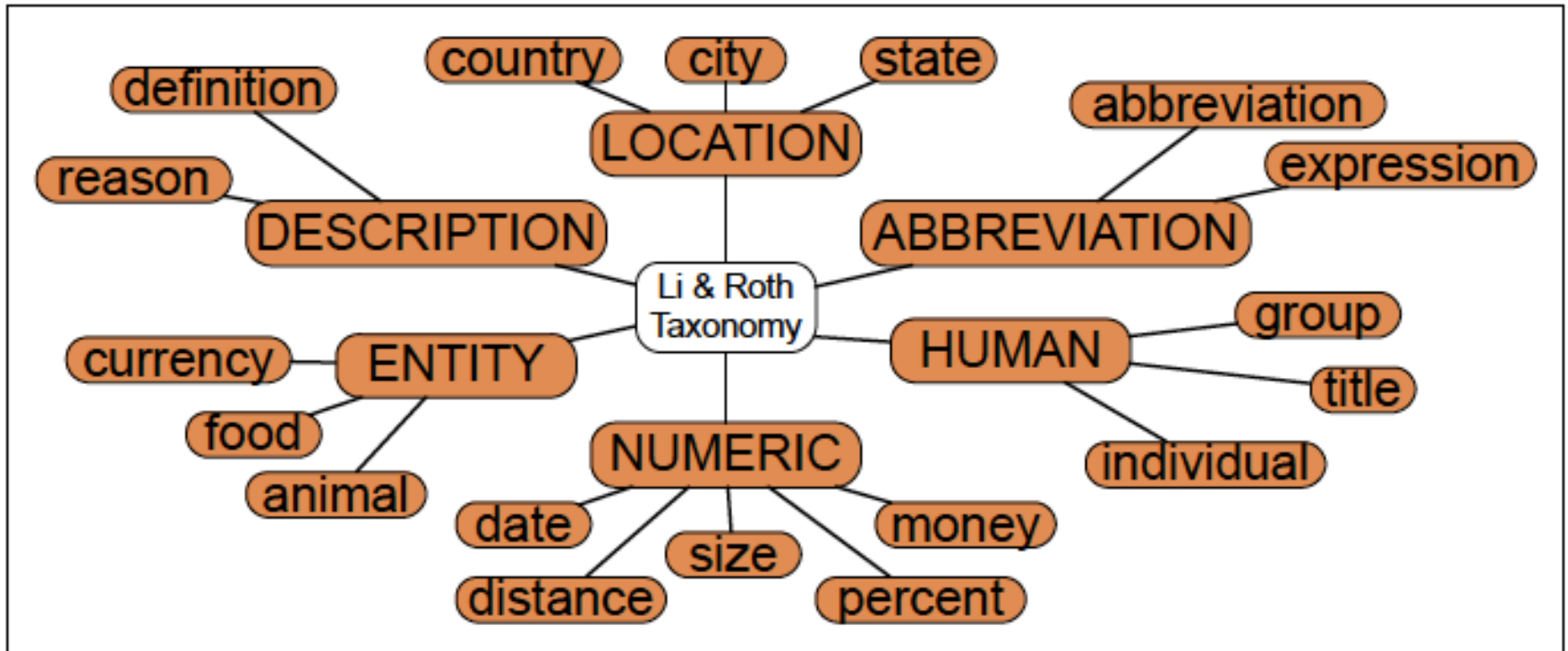


Figure 25.3 A subset of the Li and Roth (2005) answer types.

Figure from D. Jurafsky & J.H. Martin's "Speech and Language Processing", 3rd Edition.

Question processing – continued

- **Search query formulation** and/or **question reformulation**.
 - E.g., use the question's words (or **stems**), maybe **including synonyms, excluding stop-words, question words** (e.g. “who”, “when”), **focus** (since it must be replaced by the answer).
 - E.g., using hand-crafted **rules** and/or **NER tags** and/or the **parse tree** to turn the question into a statement.
 - E.g., “When was the telephone invented?” → “The telephone was invented in/at/on TIME[focus].”
 - When[focus] was NP V[passive]? → NP was V[passive] in/at/on TIME[focus].
 - Where[focus] is X? → X is in/at/on LOCATION[focus].
 - Alternatively using an **encoder-decoder model** to turn the question into a query (as in machine translation).
 - Requires suitable **parallel dataset** (with question-query pairs).

QA for document collections

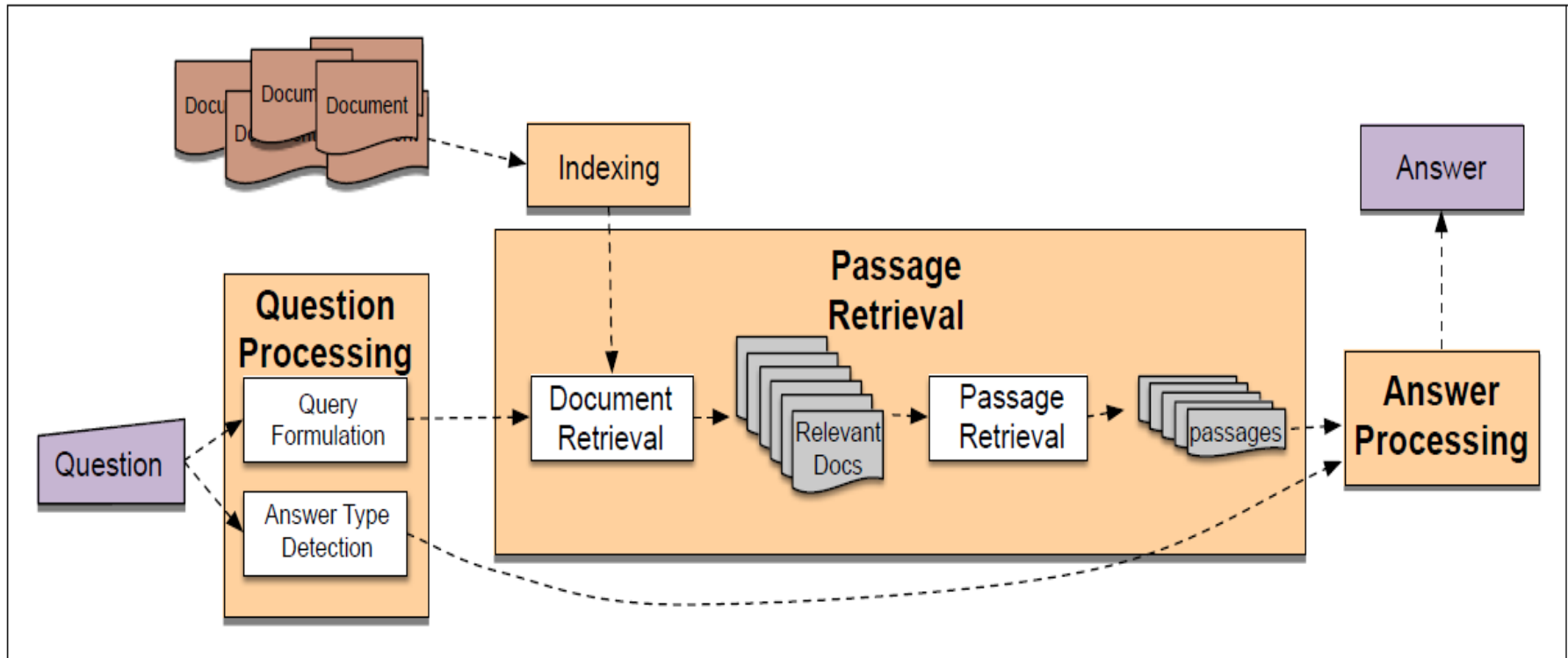


Figure 28.2 IR-based factoid question answering has three stages: question processing, passage retrieval, and answer processing.

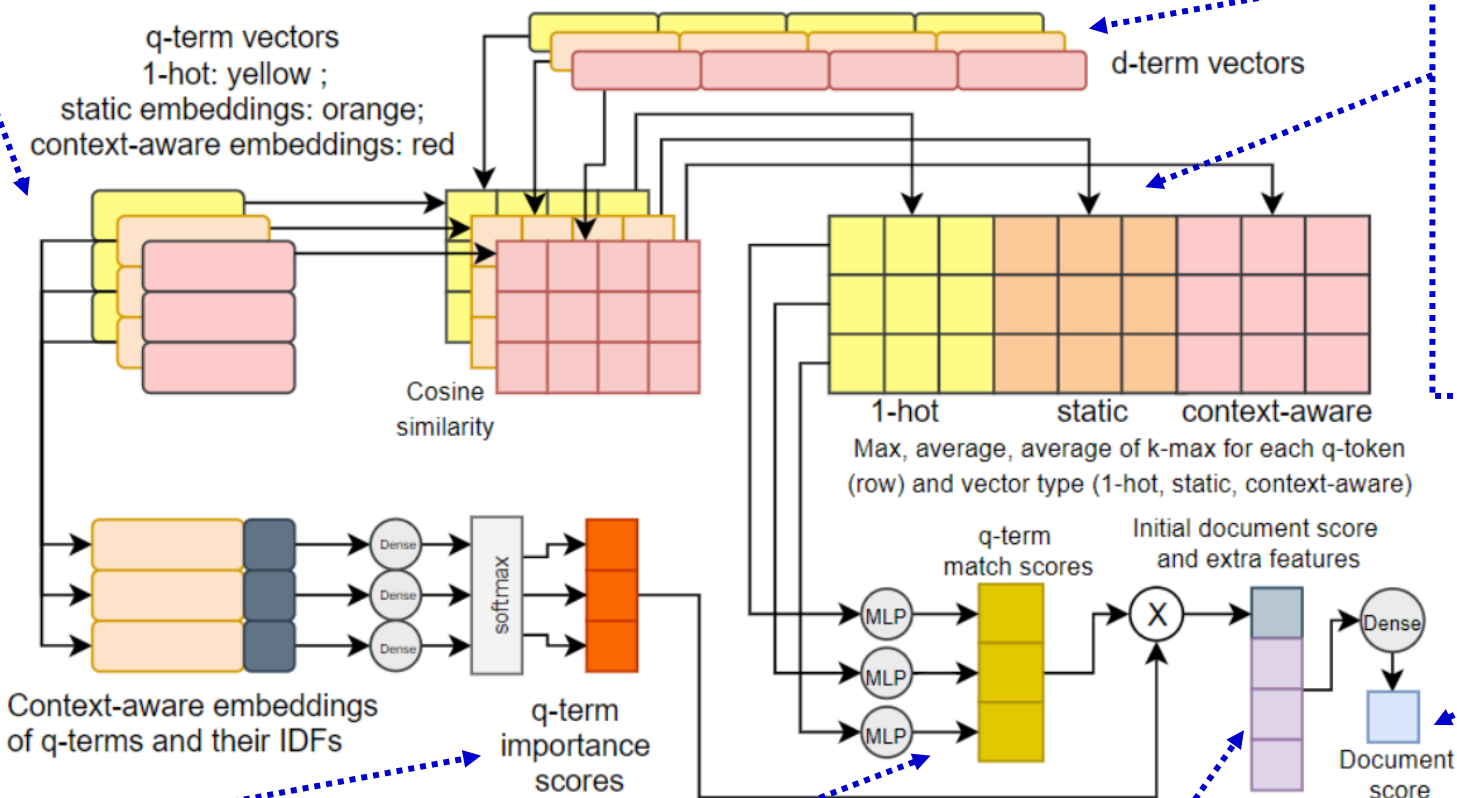
Figure from D. Jurafsky & J.H. Martin's "Speech and Language Processing", 3rd Edition.

Neural document reranking (PDRMM)

question term representations

document term representations

q-term vectors
1-hot: yellow ;
static embeddings: orange;
context-aware embeddings: red



how well each **q-term** (row) matched the document, for each view (1-hot, static embeddings, context-aware)

final **relevance score** of the document

importance of each **q-term** (extension of IDF)

how well each **q-term** term matched in the document overall

using also **traditional** document ranking features, including **BM25** score of search engine

Figure from D. Pappas et al., “AUEB at BioASQ7: Document and Snippet Retrieval”, 7th BioASQ workshop at the ECML-PKDD, 2019. http://nlp.cs.aueb.gr/pubs/aueb_at_bioasq7.pdf

Mean Average Precision (MAP)

Rank	Judgment	Precision	Recall
1	R(elevant)	1.0	0.11
2	N	0.50	0.11
3	R	0.66	0.22
4	N	0.50	0.22
5	R	0.60	0.33
6	R	0.66	0.44
...			
25	R	0.36	1.0

Average Precision (AP):

$$\frac{1}{|R|} \cdot \sum_{d \in R} \text{Precision}(d)$$

R : all relevant docs

d : doc retrieved

precision up to
the row of d

Mean Avg precision (MAP):

mean AP for all test questions

- **Ranked list of documents** returned by an IR engine.
 - Or ranked list of **passages**, or exact **answers**, ...
 - Moving to **lower rows**, **recall always increases** (when encountering a relevant doc) **or remains the same**.
 - **Precision increases** when encountering **relevant documents**, **decreases** when encountering **irrelevant documents**.

QA for document collections

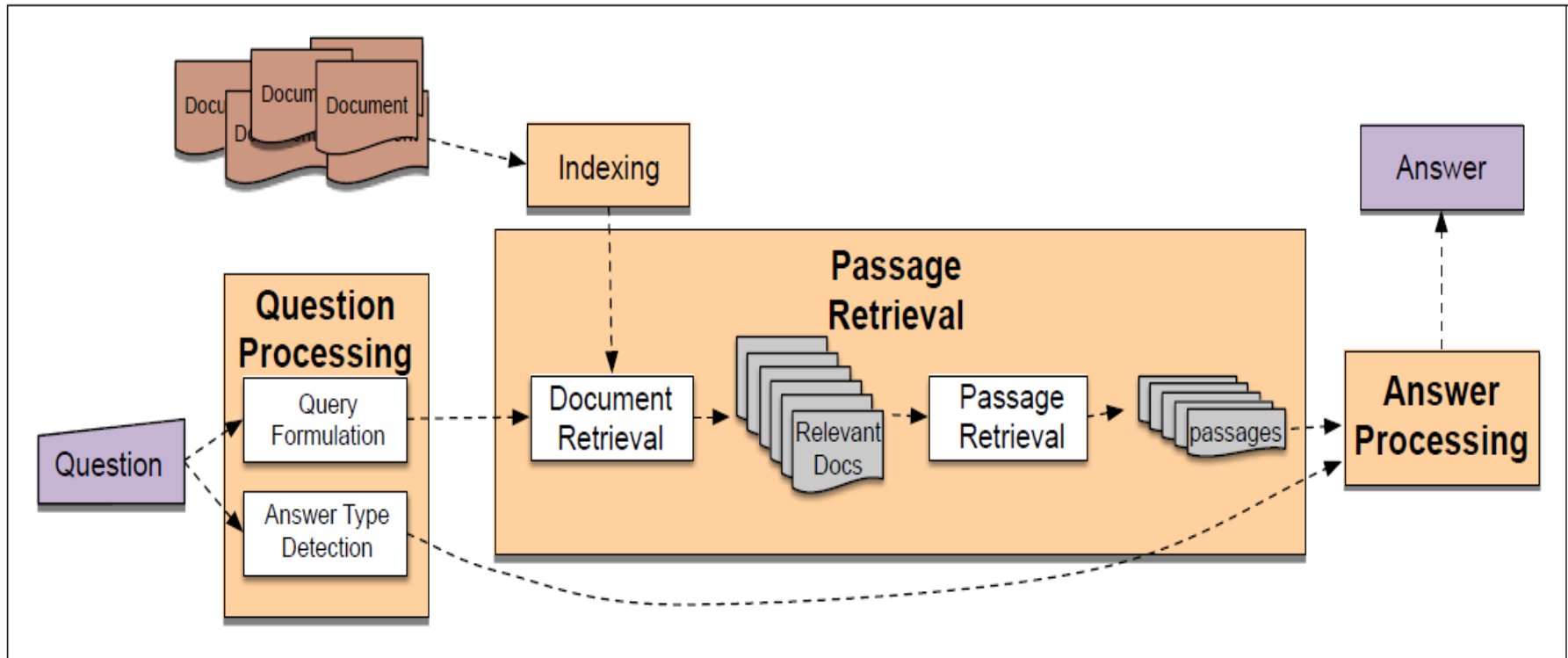


Figure 28.2 IR-based factoid question answering has three stages: question processing, passage retrieval, and answer processing.

Figure from D. Jurafsky & J.H. Martin's "Speech and Language Processing", 3rd Edition.

Passage retrieval/reranking

- The **passages (snippets)** are usually **sentences** or **paragraphs**.
 - E.g., **candidate good passages** contain **query terms** and/or **named entities** of the required **answer type**.
- **Ranking passages via machine learning**. Traditional **features**:
 - **Number of named entities** of the **right type** in the passage, **excluding** named entities of the **question**.
 - **Number and density of question keywords** in the passage.
 - **Rank of the document** that contains the passage in the **list returned by the search engine**.
 - **Question-passage similarity**, e.g., shared words/n-grams, parse tree similarity, similarity of centroids of word embeddings, ...
 - **Avg similarity of the passage** to the **other candidate passages**, e.g., to the **centroid** of their word embeddings or feature vectors.

Joint document/sentence ranking (JPDRMM)

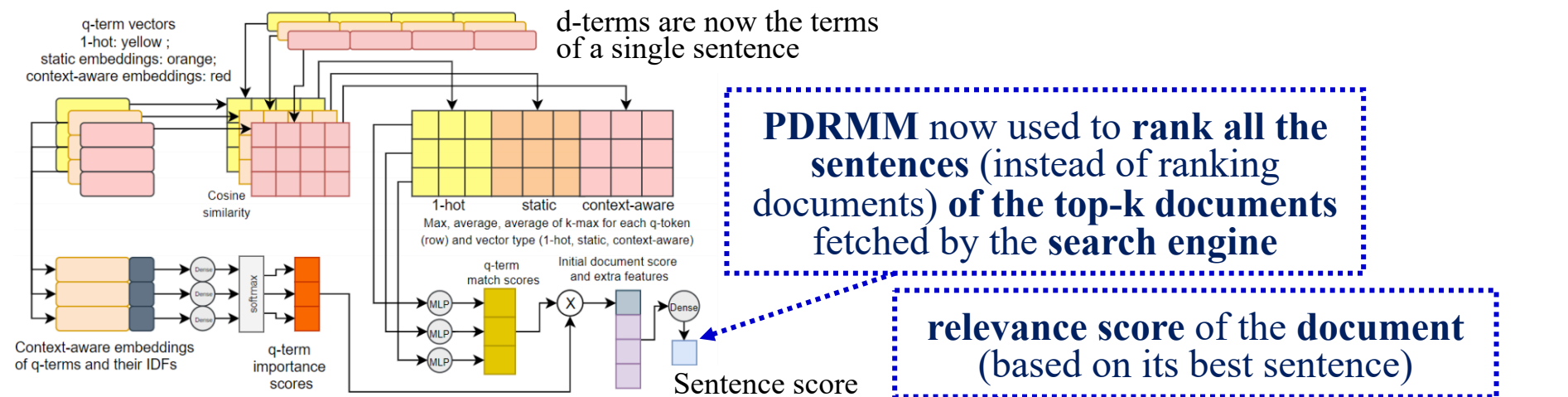


Fig. 2. PDRMM scoring documents with respect to a query. The same model can be used to score individual sentences with respect to a query, with different extra features.

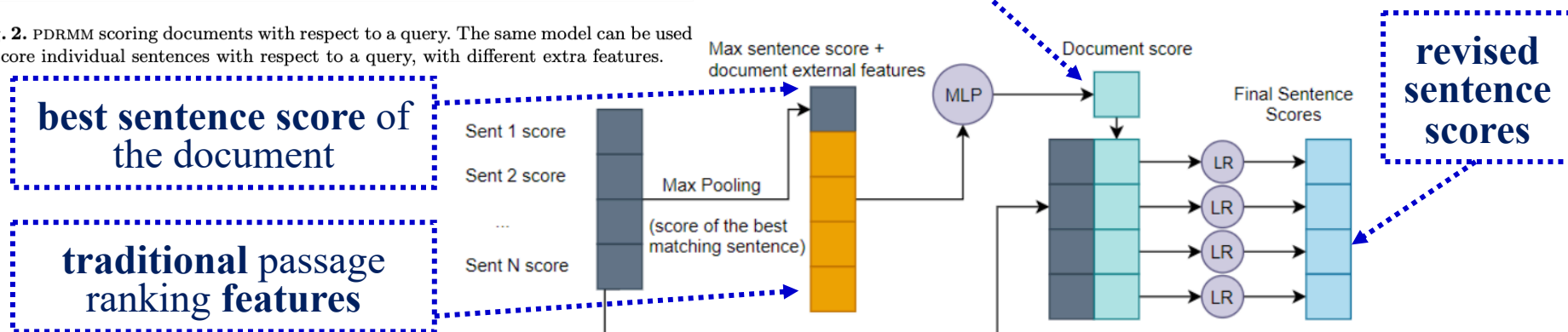


Fig. 3. The final layers of JPDRMM. The scores of the sentences (left) are generated by PDRMM (Fig. 2) operating on sentences. The maximum sentence score is concatenated with the external features of the document. An MLP produces the document score. A logistic regression layer then revises the score of each sentence, taking into account the original score of the sentence and the score of the document.

QA for document collections

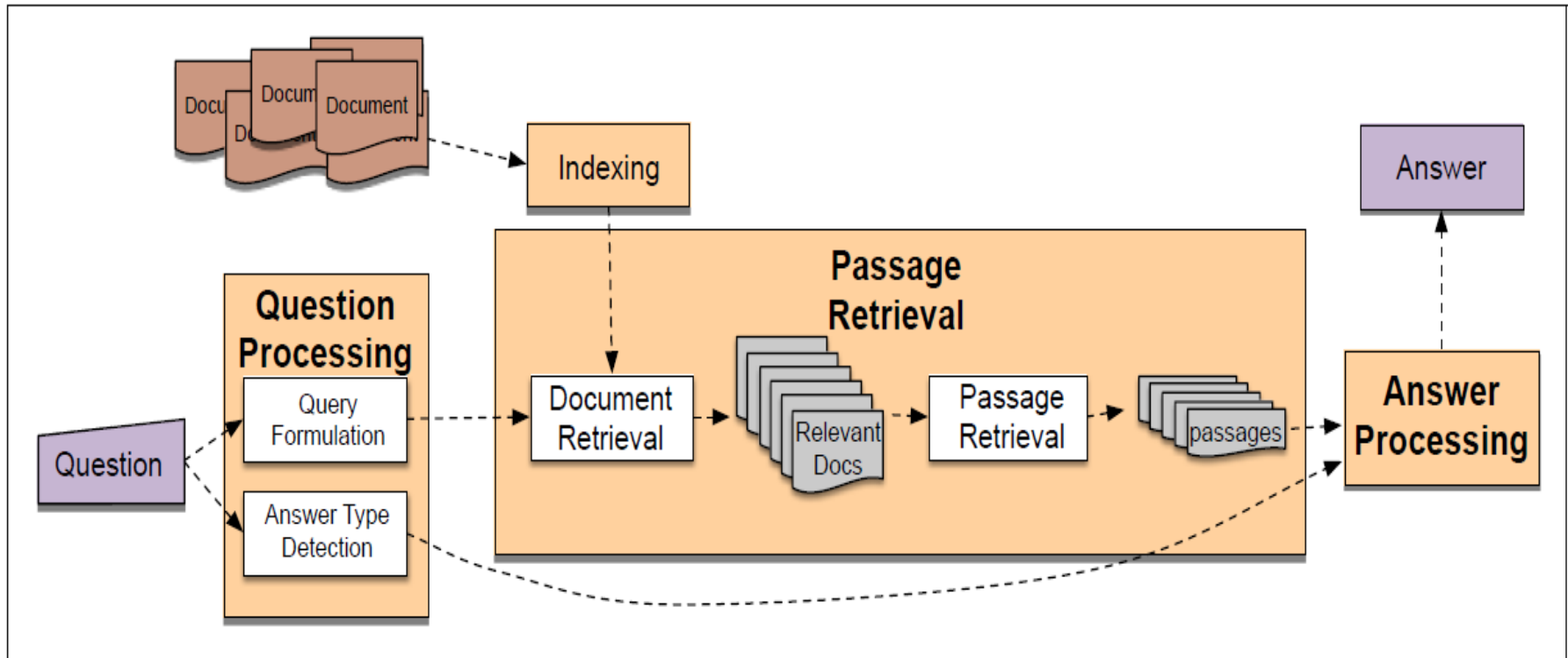


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Figure from D. Jurafsky & J.H. Martin's "Speech and Language Processing", 3rd Edition.

Exact answer extraction

- **Based on answer type and named entity recognition.**
 - If **answer type = person**, look for **person names in the best passage(s), excluding names of the question.**
 - But what if there are several person names?
 - What about **definition questions, how-to questions** etc?
- **Based on patterns and rules:**
 - E.g. if **answer type = person**, and the pattern “[*person:X*] wrote [*book:Y*]” matches the **passage**, and [*book:Y*] is a **query term**, then **return [*person:X*]**.
 - E.g., if **answer type = definition** and patterns “*X*, a kind of *Y*” or “*X* and other *Ys*” match the **passage**, and *X* is a **query term**, then **return “It is a *Y*”**.
 - **Patterns** created via **bootstrapping** (see semantics/IE slides), for **relations** (e.g., gene-disease) **common in user questions.**

Machine Reading Comprehension (MRC)

Paragraph

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called “showers”.

Question

What causes precipitation to fall?
gravity

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?
graupel

Where do water droplets collide with ice crystals to form precipitation?
within a cloud

Most MRC models try to find an exact answer in a single passage (or document).

Figure from P. Rajpurkar et al., “SQuAD: 100,000+ Questions for Machine Comprehension of Text.”, EMNLP 2016. <https://aclweb.org/anthology/D16-1264>

Neural exact answer extraction (ASReader)

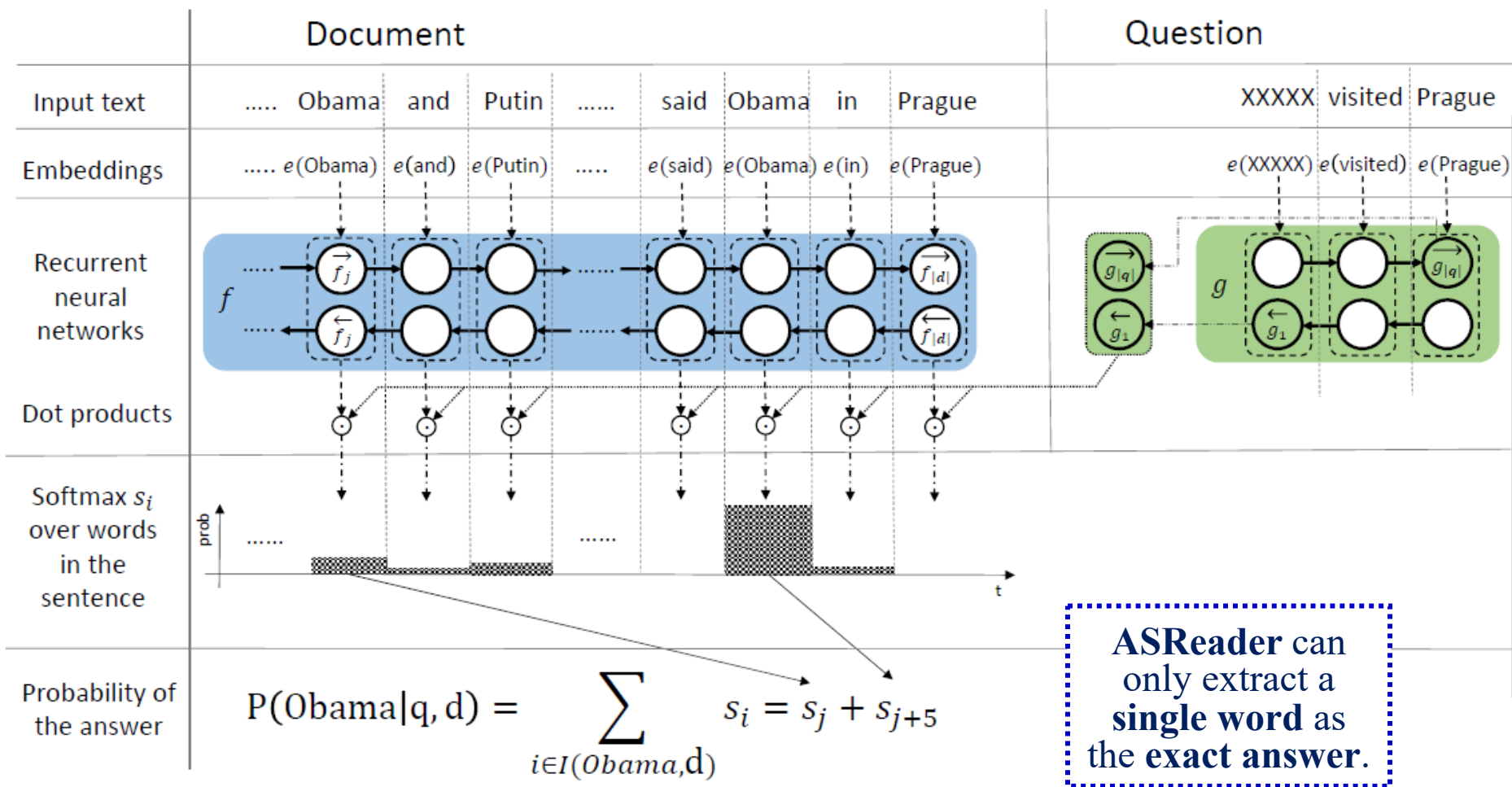


Figure from “Text Understanding with the Attention Sum Reader Network”, Kadlec et al. (2016). <http://www.aclweb.org/anthology/P16-1086>

Neural exact answer extraction (DrQA)

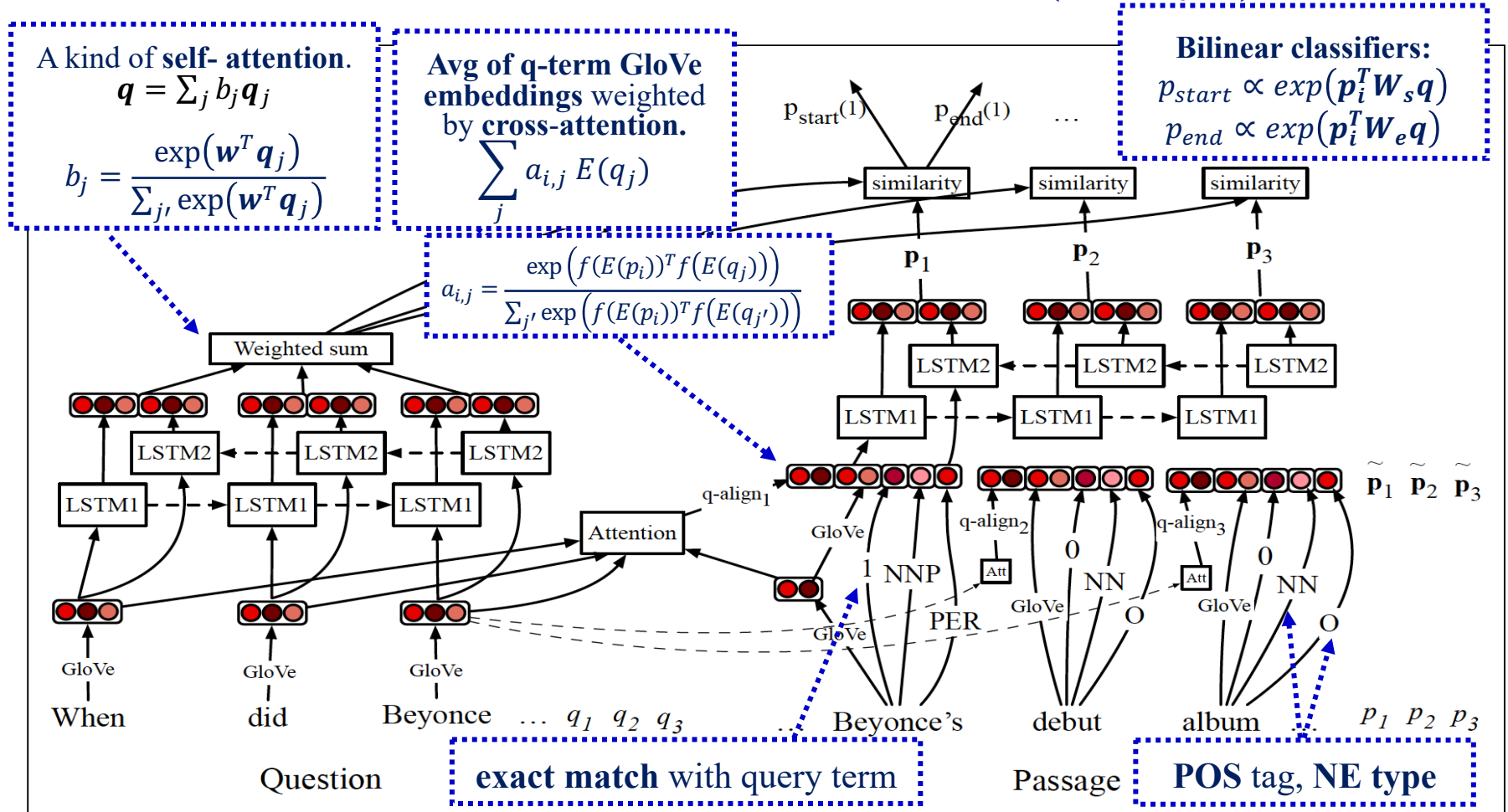


Figure 25.7 The question answering system of Chen et al. (2017), considering part of the question *When did Beyoncé release Dangerously in Love?* and the passage starting *Beyoncé's debut album, Dangerously in Love (2003).*

Figure from Jurafsky & Martin's "Speech and Language Processing", 3rd Edition. Describing the work of Chen et al. "Reading Wikipedia to answer open-domain questions", ACL 2017.

<https://www.aclweb.org/anthology/P17-1171/>

BERT – Fine-tuning for MRC

We feed the **context-aware embeddings** of the paragraph's words to a **classifier** (e.g., MLP) that classifies them as **Start-Answer, End-Answer, Other**.

If we must return a **single start-end span**, we can select the span with the **highest $p_{start} + p_{end}$** .

Starting from the **pre-trained BERT**, we **jointly train BERT (further)** and the **task-specific classifier** on (possibly few) **task-specific training examples** (paragraph-question pairs).

Most MRC models try to find an **exact answer in a *single* document** (or passage).

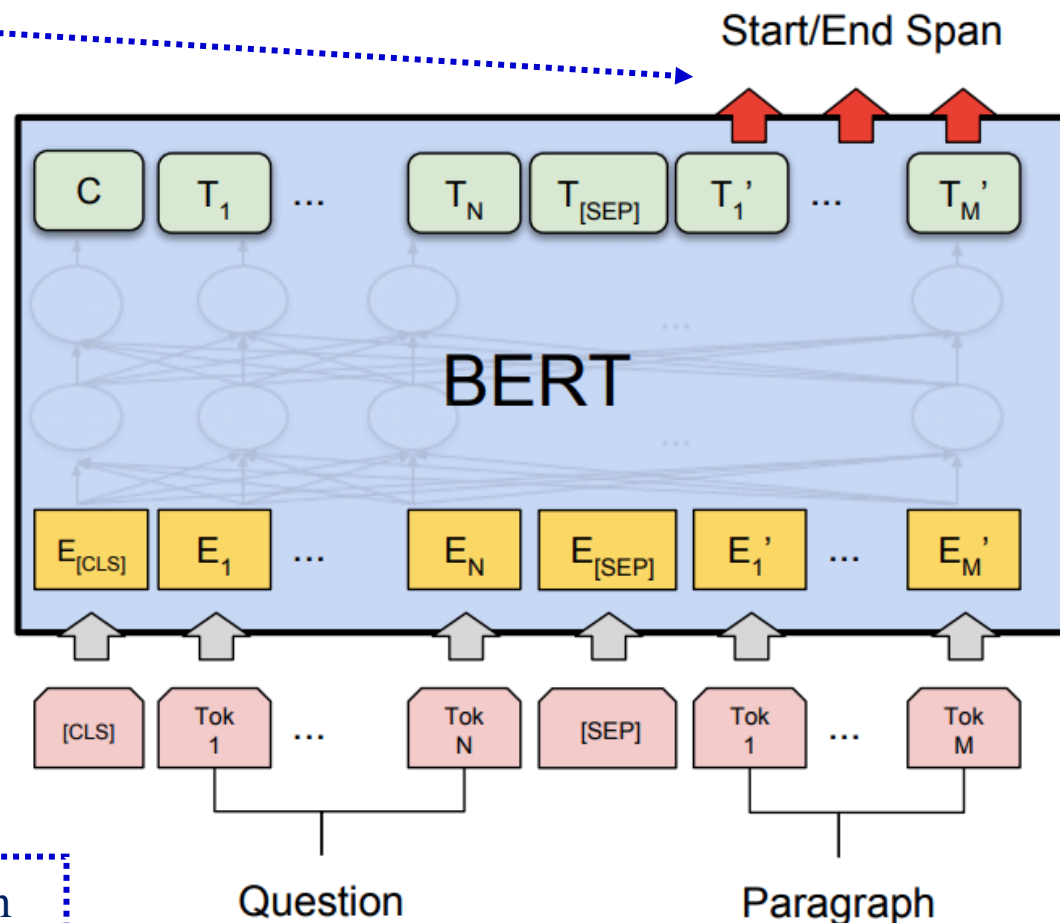


Figure from Devlin et al., “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding”, 2018. <https://arxiv.org/abs/1810.04805>

F1 score of returned spans

Treat the **span** returned by the **system** and the **gold span** as **bags of token positions** (offsets).

- E.g., system: {3rd, 4th, 5th tokens of passage}, gold: {4th, 5th, 6th tokens of passage}.
- **True Positives**: token positions in both bags.
- **False Positives**: token positions in system bag, not in gold.
- **False Negatives**: token positions in gold, not in system bag.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_{\beta} = \frac{(\beta^2 + 1) \cdot \text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}$$

Natural language interfaces to databases

The screenshot displays the Microsoft Access application window. The main window shows the 'Northwind : Database' with tabs for Tables, Queries, Forms, Reports, Macros, and Modules. The 'Queries' tab is active, showing a list of queries. One query, 'elfQ4: Which customers have placed fewer than 10 orders?', is selected and its results are displayed in a table. The table has four columns: 'Count_Of OrderID', 'Customer ID', 'Ship Name', and 'Company Name'. The results show five rows of data.

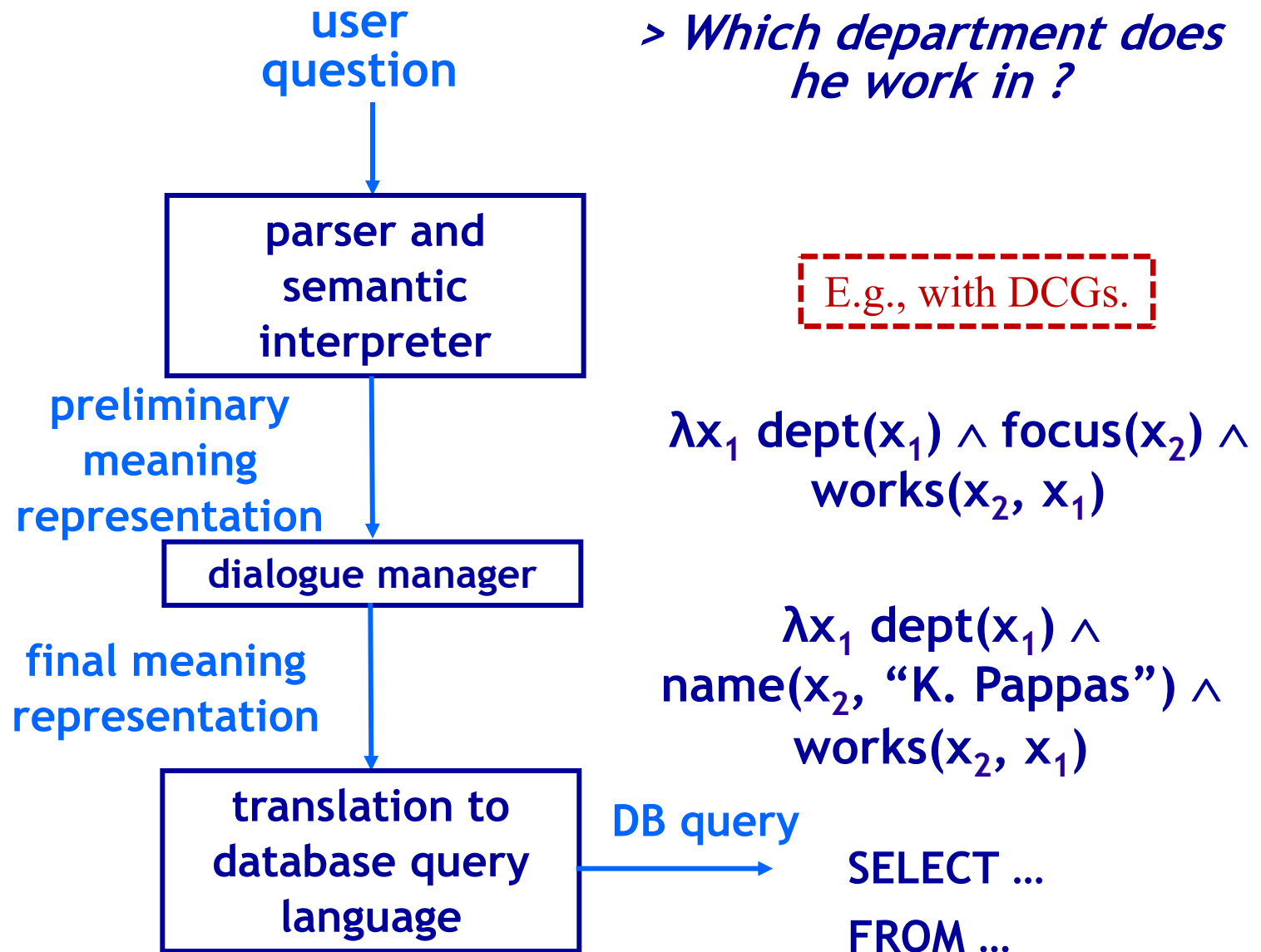
Below the query results, the 'Access ELF 97 SQL Text' window is open, showing the SQL query used to generate the results. The query is as follows:

```
SELECT DISTINCT Orders.OrderID , Customers.CustomerID FROM
Orders , Customers , Orders RIGHT JOIN Customers ON
Orders.CustomerID = Customers.CustomerID ; SELECT DISTINCT
count ( [elfQ1].OrderID ) AS [Count_Of OrderID] , [elfQ1].CustomerID
FROM [elfQ1] group by [elfQ1].CustomerID HAVING Not
IsNull([elfQ1].CustomerID) ; SELECT DISTINCT [elfQ2].* FROM [elfQ2]
WHERE [elfQ2].[Count_Of OrderID] < 10 ; SELECT DISTINCT
[elfQ3].* , Orders.ShipName , Customers.CompanyName
FROM Orders , Customers , Orders RIGHT JOIN Customers ON
Orders.CustomerID = Customers.CustomerID , [elfQ3] IN
Customers ON [elfQ3].CustomerID = Customers.CustomerID
```

The 'Access ELF Query: NORTHWIND' window is also open, showing the query name 'elfQ4: Which customers have placed fewer than 10 orders?' and the 'Respond' button.

The Windows taskbar at the bottom shows the Start button and several open applications: Exploring - ..., demos.htm ..., Paint Shop..., Delphi Ima..., myreview.h..., Natural-La..., and Microsoft Access. The system clock shows 4:29 PM.

Natural language interfaces to databases



QA for structured data

- One of the **earliest NLP applications**.
 - First systems in the 1960s and 1970s (e.g., BASEBALL, LUNAR).
- **Advantages and disadvantages**:
 - Use of natural language, **no need to learn SQL, SPARQL, ...**
 - But **often unclear** to the user **what can be asked**.
 - Often the system does not understand, **frustration...**
- It may be better to use a **controlled language**.
 - **Well defined subset of a NL**, e.g., no secondary clauses, active voice only, explicitly defined vocabulary etc.
 - But it may be **difficult** for the user to **learn** and **remember** the CL.
 - Some systems require the user to **select words from menus** that get **dynamically updated** to reflect subsequent words that can be appended to the current form of the query.
- More **appealing** combined with **speech** and/or **dialogues**.

Freebase

Discontinued, but alternatives exist (e.g., Cyc, DBPedia).
See: [https://en.wikipedia.org/wiki/Freebase_\(database\)](https://en.wikipedia.org/wiki/Freebase_(database))

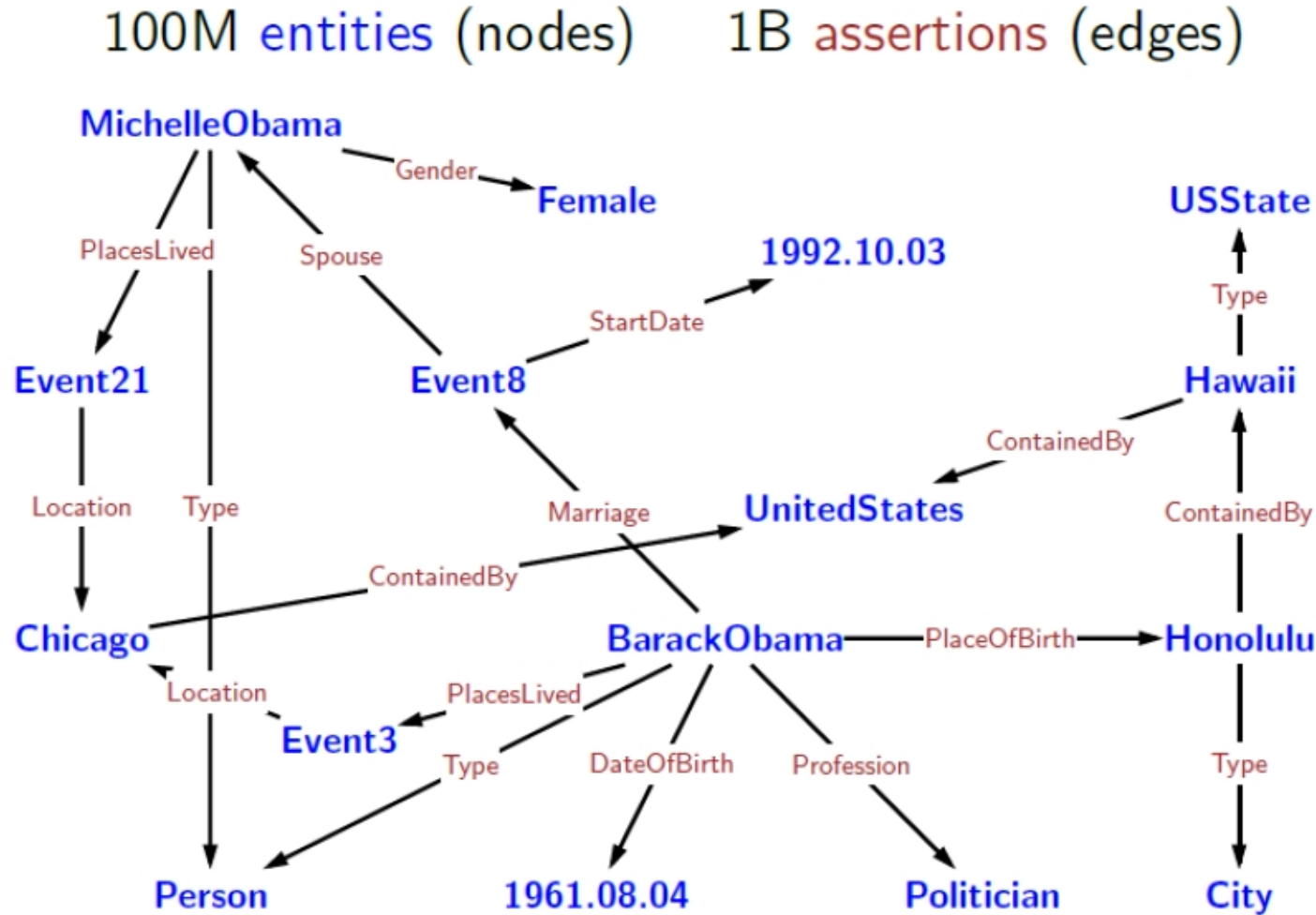


Image source: Z. Wang, S. Yan, H. Wang and X. Huang, "An Overview of Microsoft Deep QA System on Stanford WebQuestions Benchmark". Microsoft Research Technical Report MSR-TR-2014-121, 2014.

Semantic parsing via DL

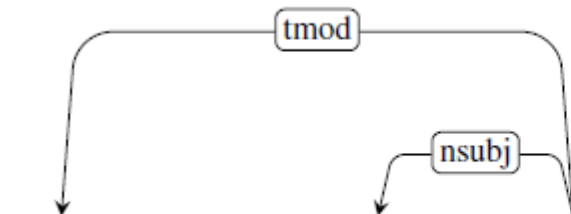


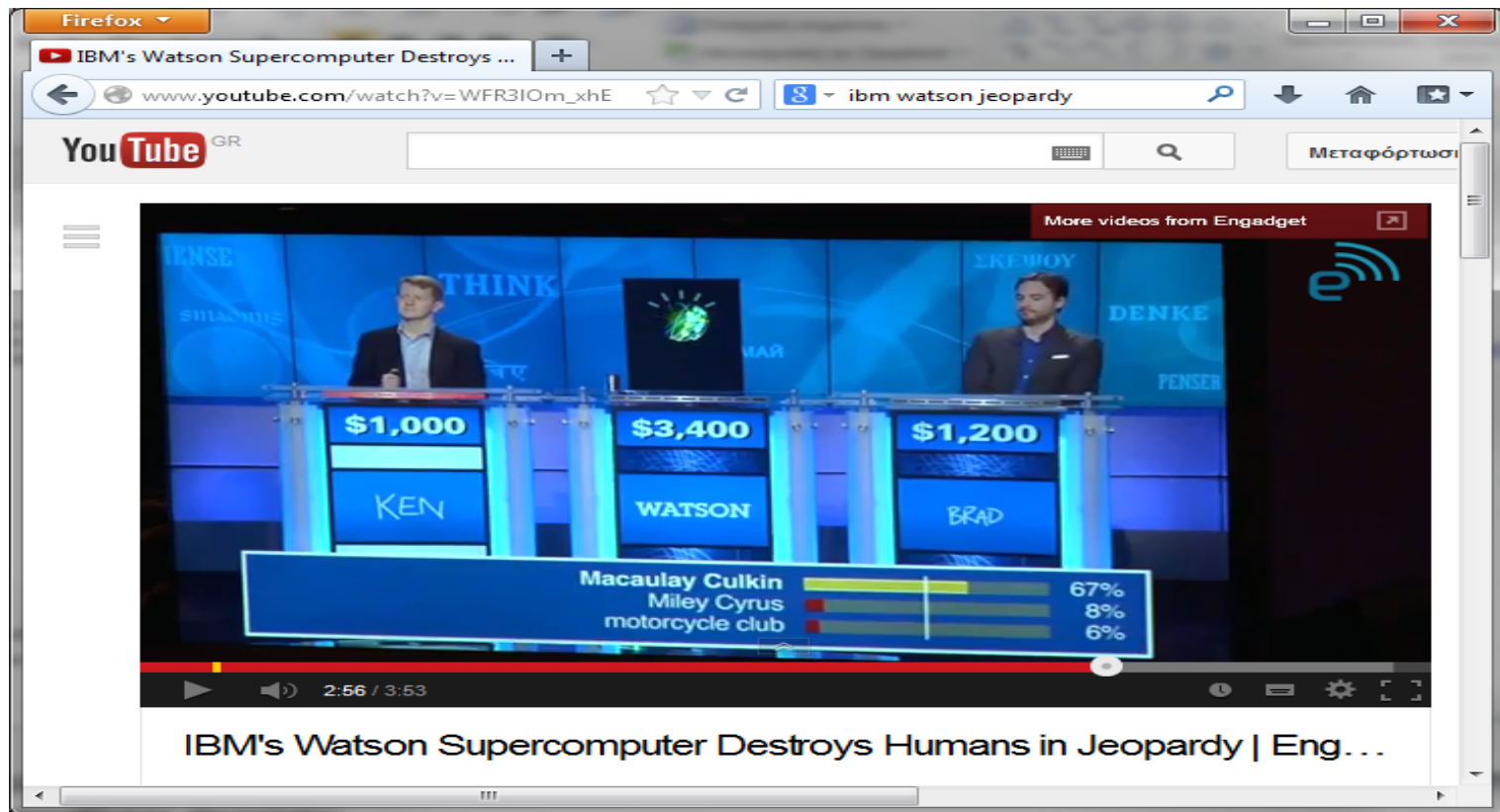
Figure from D. Jurafsky & J.H. Martin's "Speech and Language Processing", 3rd Edition.

When was Ada Lovelace born $\rightarrow$ birth-year(Ada Lovelace, ?x)

- If a dataset with **pairs of questions and semantic representations** is available, we can try to **learn the mapping**.
 - E.g., using **sequence-to-sequence models** (encoder/decoder) as in machine translation.
 - We may **encode the dependency trees** (e.g., with **Graph CNNs**) in addition to (or instead of) encoding the surface sentence.
- We can then **query the structured data** (DB, graph DB) with the **question's formula**.

IBM Watson at *Jeopardy!*

- 2011: IBM's Watson beat the best players of *Jeopardy!*
 - Queries both **structured data** and **document collections**.
 - See: http://www.youtube.com/watch?v=WFR3lOm_xhE



IBM's Watson

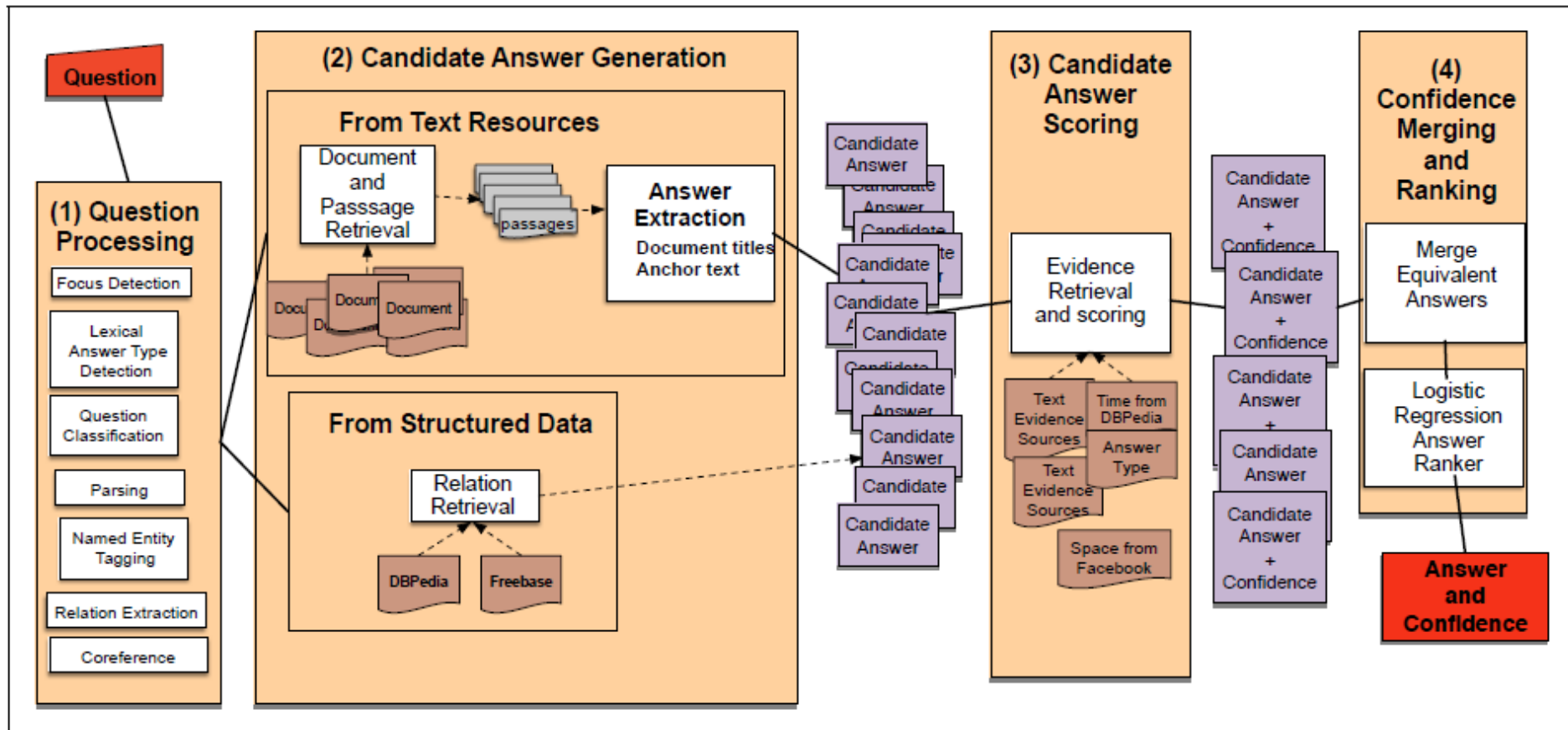


Figure 28.9 The 4 broad stages of Watson QA: (1) Question Processing, (2) Candidate Answer Generation, (3) Candidate Answer Scoring, and (4) Answer Merging and Confidence Scoring.

Figure from D. Jurafsky & J.H. Martin's "Speech and Language Processing", 3rd Edition.
Consult J&M's Chapter 25 (3rd Ed.) for further details.

Extra optional slides.

Mean Reciprocal Rank (MRR)

If a system is allowed to return up to N **answers per query**,
ranked by confidence.

- **More confident answers ranked higher.**
- **Mean Reciprocal Rank:**

$$\text{MRR} = \frac{1}{|Q|} \cdot \sum_{q \in Q'} \frac{1}{\text{rank}(q)}$$

- Q is the set of test queries, and Q' the set of test queries for which the system returned at least one correct answer.
- **For each test query $q \in Q'$, $\text{rank}(q)$ is the rank (1, 2, ..., N) of the highest correct answer returned by the system..**
- **Higher MRR for many confident (ranked high) correct answers.**
- **Unclear** what to do when the **returned answer is a span** that **partially overlaps the gold span**.

Mean Interpolated Precision (MIP@ r)

Rank	Judgment	Precision	Recall
1	R	1.0	0.11
2	N	0.50	0.11
3	R	0.66	0.22
4	N	0.50	0.22
5	R	0.60	0.33
6	R	0.66	0.44
...			
25	R	0.36	1.0

Precision _{int}	@Recall (r)
1.0	0.0
1.0	0.1
0.66	0.2
0.66	0.3
0.66	0.4
...	...
0.36	1.0

- **Interpolated precision** at each **recall level**.
 - For recall $\geq .10$, what is the best precision? Precision_{int}(0.1) = 1
 - For recall $\geq .20$, what is the best precision? Precision_{int}(0.2) = 0.66
- **Mean Interpolated Precision at Recall level r (MIP@ r):** mean Precision_{int}(r) over **all test queries**.

Mean Interpolated Precision (MIP@ r)

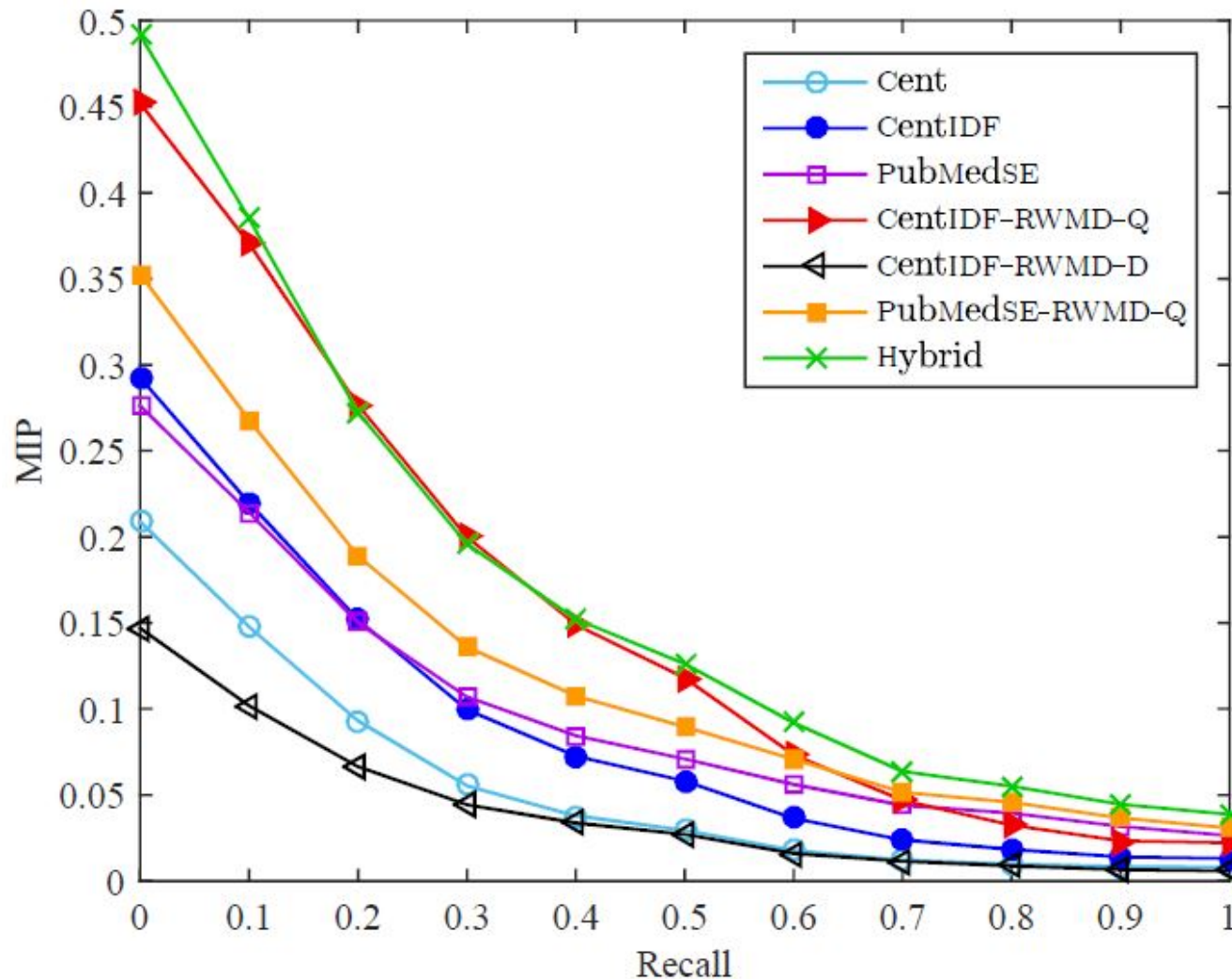


Image source: P. Malakasiotis, G. Brokos and I. Androutsopoulos, “Using Centroids of Word Embeddings and Word Mover’s Distance for Biomedical Document Retrieval in Question Answering”

(<http://nlp.cs.aueb.gr/publications.html>).

Mean Average Interpolated Precision (MAIP)

Rank	Judgment	Precision	Recall
1	R	1.0	0.11
2	N	0.50	0.11
3	R	0.66	0.22
4	N	0.50	0.22
5	R	0.60	0.33
6	R	0.66	0.44
...			
25	R	0.36	1.0

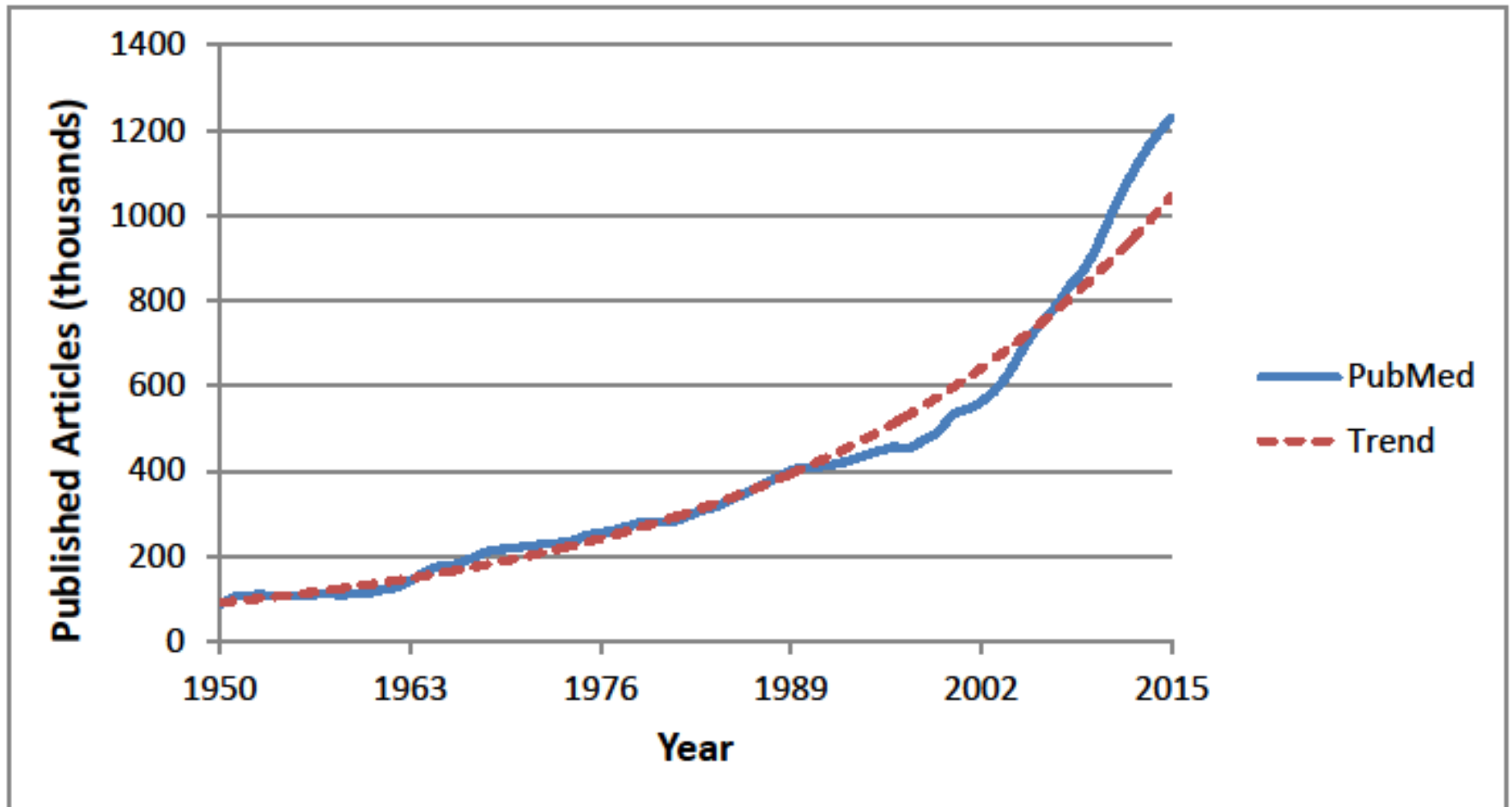
Precision _{int}	@Recall (r)
1.0	0.0
1.0	0.1
0.66	0.2
0.66	0.3
0.66	0.4
...	...
0.36	1.0

- **Average Interpolated Precision (AIP):**

$$\frac{1}{11} \cdot \sum_{r \in \{0.0, 0.1, \dots\}} \text{Precision}_{int}(r)$$

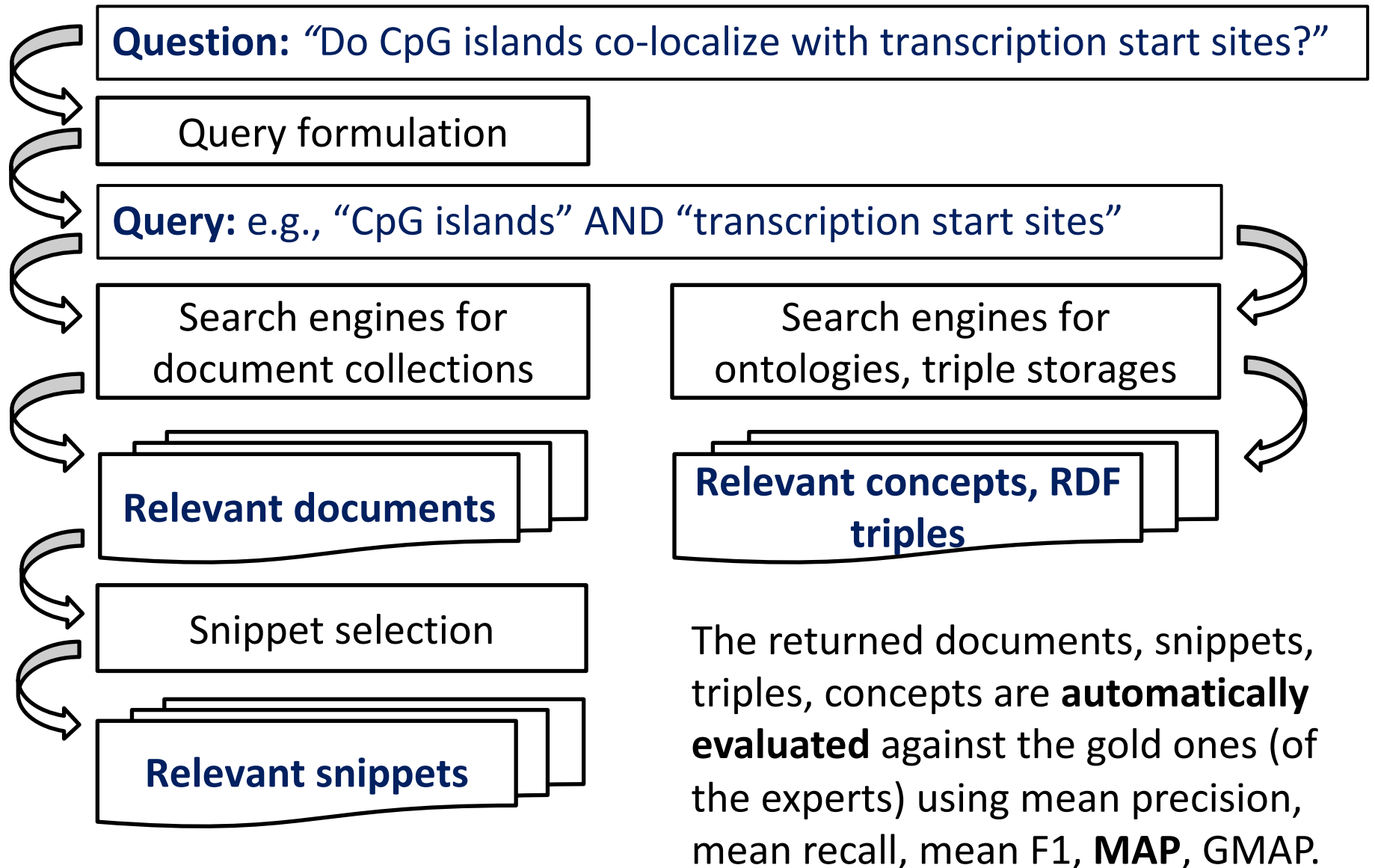
- **Mean Average Interpolated Precision (MAIP):** mean AIP over **all test queries**.

Biomedical articles per year



- In 2011: approx. 20 000 new PubMed articles *per week!*

BioASQ Task b – Phase A (Retrieval)



BioASQ Task b – Phase B (Answers)

Question: “Do CpG islands co-localize with transcription start sites?”

Relevant
documents

Relevant
snippets

Relevant concepts,
RDF triples

QA, summarization, NLG, ...

`Exact' answer: “Yes”

`Ideal' answer: “Yes. It is generally known that the presence of a CpG island around the TSS is related to the expression pattern of the gene. CGIs (CpG islands) often extend into downstream transcript regions. This provides an explanation for the observation that the exon at the 5' end of the transcript, flanked with the transcription start site, shows a remarkably higher CpG density than the downstream exons.”

BioASQ Assessment Tool

← → ↻ assess.bioasq.org/#!/ideal/536c898d2d4870f35f000001

BioA?Q Questions Ideal Answer Exact Answer Annotations Test

Which are the most under-represented oligonucleotides in higher eukaryote genomes?

✓ Save & Finalize Question Save Question

Ideal answer

TpA is the most underrepresented dinucleotide followed closely by CpG. For trinucleotides, GCA/TGC tends to be under-represented in phage, human viral, and eukaryotic sequences, and CTA/TAG is strongly under-represented in many prokaryotic, eukaryotic, and viral sequences. For higher lengths alternating Purine/Pyrimidine tracts are underrepresented up to 60%.

The oligonucleotides containing the CG and TA dinucleotide are generally under-represented in higher eukaryote genomes

We argue that the significant under-represented motif pattern of CpG in an AU context—which is found in both the ssRNA viruses and innate genes, and has decreased throughout the history of H1N1 influenza replication in humans—is immunostimulatory and has been selected against during the co-evolution of viruses and host innate immune genes

For trinucleotides, GCA.TGC tends to be under-represented in phage, human viral, and eukaryotic sequences, and CTA.TAG is strongly under-represented in many prokaryotic, eukaryotic, and viral sequences. For dinucleotides, TA is almost universally under-represented, with the exception of vertebrate mitochondrial genomes, and CG is strongly under-represented in vertebrates and in mitochondrial

For trinucleotides, GCA.TGC tends to be under-represented in phage, human viral, and eukaryotic sequences, and CTA.TAG is strongly under-represented in many prokaryotic, eukaryotic, and viral sequences. For dinucleotides, TA is almost

Information recall: ☐ ☐ ☐ ☒ ☐

Information precision: ☐ ☐ ☐ ☐ ☒

Information repetition: ☐ ☐ ☐ ☐ ☒

Readability: ☐ ☐ ☐ ☐ ☒

1 2 3 4 5

Information recall: ☐ ☐ ☐ ☐ ☐

Information precision: ☐ ☐ ☐ ☐ ☐

Information repetition: ☐ ☐ ☐ ☐ ☐

Readability: ☐ ☐ ☐ ☐ ☐

1 2 3 4 5

Information recall: ☐ ☐ ☐ ☐ ☐

Information precision: ☐ ☐ ☐ ☐ ☐

Information repetition: ☐ ☐ ☐ ☐ ☐

Readability: ☐ ☐ ☐ ☐ ☐

1 2 3 4 5

Information recall: ☐ ☐ ☐ ☐ ☐

Information precision: ☐ ☐ ☐ ☐ ☐

Ongoing work in our group to predict these....

Recommended reading

- Jurafsky & Martin (3rd Ed): Chapter 25.
 - <http://web.stanford.edu/~jurafsky/slp3/>
- For more about IR evaluation measures and IR in general, consult the book *An Introduction to Information Retrieval* by C.D. Manning, P. Raghavan and H. Schütze, Cambridge University Press, 2008.
 - Freely available at <http://nlp.stanford.edu/IR-book/information-retrieval-book.html>

